



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Fifth Semester- Bachelor of Technology (Electrical Engineering) Session 2020-21

SPECIALIZATION – Electrical Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam	Mid sem Tests	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EX-51	Electrical Machine-II	3		2	4	70	20	10	30	20	150
2	EX-52	Electrical Power Generation & Economy	3	1	2	5	70	20	10	30	20	150
3	EX-53	Power Electronics	3	1	2	5	70	20	10	30	20	150
4	EX-54	Electrical & Electronics material	3	1		4	70	20	10			100
5	EX-55	Seminar	-	-	6	3	-	-	-	-	100	100
6	EX-56	Minor project-I	-	-	6	3	-	-	-	-	100	100
7		Industrial training	To be completed anytime during Fifth/Sixth semester. Its evaluation/credit to be added in Seventh Semester.									
		TOTAL	12	3	18	24	280	80	40	90	260	750

Theory Credits: 15
 Practical Credits: 09
 Total Credits: 24

Theory Marks: 280
 Practical Marks: 90
 Quiz, Mid Term, Lab. Work: 350

Total: 750



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

**SYLLABUS OF COURSES TO BE OFFERED
“Bachelor of Technology”**

Disclaimer: The Undergraduate programme in Bachelor of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

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Bachelor of Technology
V
Semester
Electrical & Electronics
Engineering (EEE)

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

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Bachelor of Technology

Branch - Electrical & Electronics Engineering

EX-51 Electrical Machine-II

Unit-I

D.C. Machine-I Basic construction of DC machines; types of DC machines and method of excitation; lap and wave windings; Emf equation; armature reaction and methods of limiting armature reaction; Commutation process and methods for improving commutation; Basic performance of DC generators and their performance characteristics; Metadyne and Amplidyne; permanent magnet DC motors; Brush less dc motors.

Unit-II

D.C. Machine-II Basic operation of DC motors; Torque equation; Operating characteristics of DC motors, Starting of DC motors- 2point, 3 point and 4 point starters; speed control of DC motors; losses and efficiency of DC machines; testing of DC machines, direct testing, Swinburne's test and Hopkinson's test. Application of DC machines.

Unit-III

Synchronous Machine-I Construction; types of prime movers; excitation system including brushless excitation; polyphase distributive winding, integral slot and fractional slot windings; emf equation, generation of harmonics and their elimination; armature reaction; synchronous reactance and impedance, equivalent circuit of alternator, relation between generated voltage and terminal voltage, voltage regulation of alternators using synchronous impedance, mmf, zpf and new A.S.A method.

Unit-IV

Synchronous Machine-II Salient pole machines; two reaction theory equivalent circuit model and phasor diagram; determination of X_d and X_q by slip test; SCR and its significance; regulation of salient pole alternator, power angle equation and characteristics; synchronizing of alternator with infinite busbar,; parallel operation and load sharing; synchronizing current, synchronizing power and synchronising torque coefficient; synchro scopes and phase sequence indicator; effect of varying excitation and mechanical torque.

Unit-V

Synchronous machine-III Synchronous motor operation, starting and stopping of synchronous motor, pull in torque, motor under load power and torque, reluctance torque, effect of excitation, effect of armature reaction, power factor adjustment, V curves, inverted V curves, synchronous motors as power factor correcting device, super synchronous and sub synchronous motors, hunting and damper winding efficiency and losses. Analysis of short circuit oscillogram, determination of various transient, sub transient and steady reactances and time constants, expression of transient and sub transient reactances in terms of self and mutual inductances of various winding, short circuit current, equivalent circuit. Single phase synchronous motors- hysteresis motor, reluctance motor. Repulsion motor, stepper motor, switched reluctance

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REFERENCE BOOKS:-

1. M.G. Say, Performance & design of AC machines, CBS publishers & distributors, Delhi, 3rd edition
2. P.S. Bhimbra, Electrical Machinery, Khanna Pub.
3. P.S. Bhimbra, Generalized theory of Electrical Machines, Khanna publishers, Delhi,
4. Ashfaq Husain, Electric Machines, Dhanpat Rai, New Delhi
5. I.J. Nagrath & D.P. Kothari, Electric Machines, Tata McGraw Hill, New Delhi.

EXPERIMENT LIST:-

1. To plot magnetisation characteristic of a separately excited DC generator.
2. To perform load test on DC generators
3. To perform load test on DC series and shunt motor
4. To perform Swinburn's test on a DC machine and find out its efficiency under full load condition.
5. To conduct Hopkinson's test on a pair of DC shunt machine
6. To perform OCC and SCC test on an alternator and determine its regulation.
7. To determine regulation of alternator using mmf and zpf methods.
8. To synchronise alternator with infinite bus bar.

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EX- 52 Electrical Power Generation & Economy

Unit-I

Introduction: Energy sources and their availability, Principle types of power plants, their special features and applications, Present status and future trends. Hydro Electric Power Plants: Essentials, Classifications, Hydroelectric survey, Rainfall run-off, Hydrograph, Flow duration curve, Mass curve, Storage capacity, Site selection, Plant layout, various components, Types of turbines, Governor and speed regulation, Pumped storage, Small scale electric plants (mini and micro).

Unit-II

Thermal Power Plant: General developing trends, Essentials, Plant layout, Coal Preparation, Handling, Feeding and burning, Cooling towers, Ash handling, Water treatment plant, High pressure boilers and steam turbines, Components of thermal power plant.

Unit-III

Non-Conventional Power Generation: Geothermal power plants, Electricity from biomass, Direct energy conversion systems (Solar and Wind), Thermo-electric conversion system, Fuel cells, Magneto-Hydro dynamic system.

Unit-IV

Gas Turbine Power Plants: Field of use, Components, Plant layout, Comparison with steam power plants, combined steam and gas power plants. Nuclear Power Plant: Nuclear fuels, Nuclear energy, Main components of nuclear power plant, Nuclear reactors types and applications, Radiation shielding, Radioactive and waste disposal safety aspect.

Unit-V

Power Plant Economics: Cost of electrical energy, Selection of type of generation and generation equipment, Performance and operating characteristics of power plants, Economic scheduling principle, Load curves, Effect of load on power plant design, Load forecasting, electric tariffs, Peak load pricing.

REFERENCE BOOKS:-

1. Deshpande, M.V., Power Plant Engineering, Tata McGraw Hill (2004).
2. Gupta, B.R., Generation of Electrical Energy, S. Chand (1998)
3. Deshpande, M.V., Electrical Power System Design, McGraw Hill (2004). Wood, A.J. and Wollenberg, B.F., Power Generation and Control, John Wiley (2004).

EXPERIMENT LIST:-

1. Study of Steam Power Plant (Thermal Power Plant).
- 2 Study of Hydro-Electric Power Station.
- 3.Study of Nuclear Power Plant.
1. Study of Diesel Power Plant.
2. Measurements of the voltage and current of the solar cells by changing the direction of solar panel

Bachelor of Technology

Branch - Electrical & Electronics Engineering

EX -53 Power Electronics

UNIT-I

Advantages and application of power electronic devices characteristics, Symbol & application of power diodes, power transistors, GTO, Triac, Diac, Power MOSFET, IGBT, LASCR, Fast recovery diode, schottky diode MCTs. Principle of operation of SCR, Two transistor analogy, brief idea of construction of SCR, Static characteristics of SCR, Condition of turn on & off of SCR Gate characteristics, Method for turning on of SCR, Turnoff methods, different commutation techniques (Class A,B,C,D,E, & F Commutation) firing of SCR, Resistance firing Ckt, Resistance capacitance firing circuit, UJT firing cut, and ramp triggering, firing for 3- Φ circuit. SCR rating & protection of SCR over voltage, Over current, Superior firing, Design of snubber circuit and protection of gate of SCR, heating, cooling & mounting of SCR series and parallel operation of SCR, String efficiency & problem associated with series and parallel operation of SCR.

UNIT-II

Operation and analysis of single phase (Half wave & Full Wave) and multiphase (Three Phase) uncontrolled and controlled rectifier circuit with resistive, resistive & inductive load (continuous & non continuous conduction, FW small & very large inductive loads) and RLE loads. Estimation of average load voltage and load current for above rectifier circuits active and reactive power input. Effect of freewheeling diode and source inductance on performance of these rectifier circuits . Comparison of mid-point & Bridge rectifier circuits

UNIT-III

Series and parallel inverter, Voltage source & current source inverter, Single phase and three phase bridge inverter, Self-cumulated inverters,, Mc- murray& MC murray bed ford inverters, Voltage control of single phase and three phase bridge inverter, Harmonics & their reduction techniques.

UNIT-IV

Principle of chopper operation, Various control strategies in chopper, Step up & step-up/step down choppers, chopper configuration (Type A,B, C,D, & E), Steady state analysis of chopper circuits, Current & voltage commutation of chopper circuits Jones & Morgens chopper.

UNIT-V

Single phase (midpoint & bridge configuration) and three phase cyclo convertor configuration and operating principles. AC voltage controllers (using SCRs & Traics) single phase full wave controller with R and RL load, Estimation of RMS load voltage, RMS load current and input power factor, three phase AC voltage controller (Without analysis) Dual converter Switched mode voltage regulator buck, Boost, Buck & Boost, Ck regulators.

REFERENCE BOOKS:-

- 1- M.H. Rashid, Power Electronics Circuits, Devices and Applications, Pearson Education, Singapore, 1993.
- 2- M Ramsmoorthy, An Introduction to transistor and their application, Affiliated EastWest Press
- 3- P.C. Sen, Power Electronics, TMH
- 4- M.D. Singh, K.B. Khanchandani, Power Electronics, TMH, Delhi, 2001.
- 5- Chakravarti A., Fundamental of Power Electronics and Drives, Dhanpat Ray & Co.
- 6- Dr. P.S. Bhimbhra, Power Electronics, Khanna Pub. Vedam Subramanyam, Power Electronics New Age International Revised II ed. 2006.

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EXPERIMENT LIST:-

- 1- VI Characteristics Of SCR.
- 2- VI Characteristics Of DIAC
- 3- VI Characteristics Of BJT.
- 4- Characteristics Of TRIAC.
- 5- VI Characteristics Of MOSFET.
- 6- Transfer Characteristics Of MOSFET.
- 7- Output Characteristics Of IGBT.
- 8- Transfer Characteristics Of IGBT.
- 9- Single Phase SCR Half Controlled Converter With R Load
- 10- 1 ϕ Scr Fully Controlled Converter With R-Load

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EX- 54 Electrical and Electronic Materials

Unit I

Crystal structure of materials, crystal systems, unit cells and space lattices and defects, Classes of Engineering Materials – Metals & alloys, ferrous and non-ferrous alloys, low alloy steels, aluminium alloys, copper alloys, stainless steels, cast iron, ceramics, organic polymers and composite material. Classification of solids from electrical engineering point of view. Conducting material – properties of conductors, characteristics of good conductor material, commonly used conducting materials, conductor materials for overhead lines, types of conductors, conductor for underground cables, conductor materials used in electrical machines, resistor materials, types of resistors, materials for bus bar.

Unit II

Dielectric Materials: Dielectric strength, factors affecting dielectric strength, dielectric loss, dissipation factor, factors affecting dielectric loss, permittivity & polarization, charging and discharging of dielectric, conduction through dielectric. Application of dielectric, different types of capacitors and materials used for them. Insulating materials, their– thermal and chemical, mechanical & electrical property. Insulating materials like ceramic, mica, glass, rubber, resins, wax varnishes, Class of insulator. Transformer oils & their testing. Piezoelectricity & Ferro electricity

Unit III

Concept of energy band diagram for materials - conductors, semiconductors and insulators Applications of semi conductor materials: type of semi conductors, working and applications of semiconductors, Temperature sensitive elements, photoconductive cells, photo voltaic cells; Varistor, Hall effect generator, LCD, Light dependent resistors, LEDs, piezo – electric materials, semiconductor laser and its characteristics, photo conductors – photo diodes, avalanche photo diode, photo transistors.

Unit IV

Classification of magnetic materials: Dia-magnetism, Para magnetism, Ferromagnetism, magnetization curve, hysteresis loop, Magnetostriction, Factors affecting permeability and hysteresis, Anti – ferromagnetism, Ferromagnetism, Magnetic resonance, B-H curve for different magnetic materials, loss of magnetism, impurities in ferromagnetic materials, soft and hard magnetic materials, ferrites

Unit V

Superconductivity & it's application. Materials of MHD generator, Thermoelectric generators, Thermionic conductors, Physical properties & Electrical properties of SF₆, Specification of SF₆ gas for GIS application, Advantages and Applications of SF₆, Nanomaterials, Ultra Light materials and metallic foams.

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REFERENCE BOOKS:-

1. A.J. Dekker; Electrical Engineering Materials; PHI
2. William F Smith, JavadHashemi, Ravi Prakash 'Material science and engineering', McGraw Hill.
3. James F. Shackelford, Madanapalli K. Muralidhara 'Introduction to Materials Science for Engineers', Pearson
4. Ian P. Jones ' Materials Science for Electrical and Electronics Engineers' Oxford university press
5. C. S. Indulkar and S. Thruvengadem; Electrical Engineering Materials; S. Chand.
6. TTTI Madras; Electrical Engineering Materials; TMH

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EX-56 Minor project-I

Students have to complete minor project by using any software or hardware.
Technically it should be implemented for skill development of the students.

